

## EDUCATION

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- **Rochester Institute of Technology (RIT)** Rochester, NY  
*Master of Science in Data Science; GPA: 3.9* Aug. 2023 – May 2026

## EXPERIENCE

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- **Morningstar** Mumbai, India  
*Associate Software Engineer* May 2021 – May 2023
  - **End-to-End Product Ownership:** Built portfolio analytics tools with full ownership from data pipeline to user-facing interface in Vue.js, improving report clarity for 100k+ users.
  - **Scalable Content Platform:** Designed CMS-driven FAQ system using Contentstack APIs, enabling non-technical stakeholders to manage documentation independently.

## RESEARCH

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- **Dual-Objective Neural Architecture Search Ablation Study** Rochester, NY  
*Student ML Researcher, RIT Evolutionary Computing Lab* Jan. 2025 – Present
  - **C++ Distributed Implementation:** Refactored C++ `SpeciationStrategy` classes to track evaluating genomes across MPI worker nodes, resolving async bottlenecks; orchestrated 50+ parallel ablation jobs on a Slurm-managed 2,304-core HPC cluster isolating effects of SWEET selection vs. Harada pruning on simultaneous MSE and size reduction.
  - **Statistical Validation:** Applied Mann-Whitney U tests ( $p < 0.05$ ) across 4 time-series datasets to validate significance of dual-objective compression and accuracy gains.
- **Singh, A., Lyu, Z., Desell, T.** “Biologically-Inspired Homeostasis for Neuroevolution.” *EvoStar 2026* - 23% model compression ( $p = 0.0019$ ) without MSE degradation on multivariate time-series forecasting.

## PROJECTS

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- **Transformer Fine-Tuning for Text Summarization**  
*PyTorch, HuggingFace Transformers, PEGASUS/BART/T5, Python*
  - **LLM Fine-Tuning:** Fine-tuned Transformer models (PEGASUS, BART, T5) on the SAMSum dialogue dataset using HuggingFace PEFT ecosystem moving beyond pre-trained weights to task-specific model adaptation for generative summarization.
  - **Modular NLP Pipeline:** Built production-style modular pipeline covering data ingestion, tokenization, training with checkpoint management, and evaluation demonstrating software engineering discipline alongside ML experimentation.
- **Generative AI Video Compliance & Brand Safety System**  
*LangGraph, Azure Video Indexer, Azure AI Search, LangSmith*
  - **LLM-Based Content Quality Assessment:** Leveraged LLMs to detect policy violations and assess content quality across video and text modalities applicable to safe and high-quality GenAI model deployment.
  - **LLMOps & Evaluation:** Integrated LangSmith tracing and Azure Monitor for prompt evaluation and latency monitoring, supporting iterative model quality improvement in production.

## TECHNICAL SKILLS

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- **Languages:** Python, C++, SQL, JavaScript, Bash
- **ML & AI:** PyTorch, TensorFlow, Transformers (HuggingFace), LLM Fine-Tuning (PEFT/LoRA), Collaborative Filtering, Neural Architecture Search, Distributed Training (MPI/Slurm/HPC), XGBoost, Scikit-learn, RAG, LangChain, LangGraph, LLMOps
- **Infrastructure & Data:** FastAPI, Docker, Kubernetes, GCP, AWS, Azure (OpenAI, AI Search, Monitor), PostgreSQL, Pinecone, Pandas, NumPy, Airflow, Git